

ECE250: Signals & Systems Quiz-7

Date: 29th October, 2025

Total Marks: 10

Duration: 15 Minutes

Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
 2. Institute Plagiarism policy are strictly applicable.
-

[CO1 CO2 CO4] [Q1]: A causal and stable LTI system S has the frequency response

$$H(j\omega) = \frac{j\omega + 4}{6 - \omega^2 + 5j\omega}$$

- (a) Determine a differential equation relating the input $x(t)$ and output $y(t)$ of S. **[3 Marks]**
- (b) Determine the impulse response $h(t)$ of S. **[3 Marks]**
- (c) What is the output of S when the input is **[4 Marks]**

$$x(t) = e^{-4t}u(t) - t^2e^{-4t}u(t)?$$